

Internet and How It Works

Introduction:

Welcome, everyone! Today, we're going to dive into the fascinating world of the Internet. By the end of this lecture, you'll understand what the Internet is, how it functions, and the key components that make it all work. Let's get started!

1. What is Wi-Fi?

Wi-Fi is a facility allowing computers, smartphones, or other devices to connect to the internet or communicate with one another wirelessly within a particular area.

2. What is the Internet?

Imagine a gigantic web that spans the entire globe, connecting millions of computers. This web is the Internet. It allows computers to communicate and share information with each other, no matter where they are located. The Internet is the backbone of our digital world, enabling us to access websites, send emails, stream videos, and much more.

3. Internet Protocols: The Rules of Communication

For computers to communicate effectively, they need to speak the same language. This is where **internet protocols** come in. Internet protocols are a set of rules that determine how data is sent and received over the Internet. The two main protocols are:

- **TCP (Transmission Control Protocol):** Ensures that data is sent and received accurately. Think of it as a reliable mail service that makes sure your packages (data) arrive safely and in order.
- **IP (Internet Protocol):** Directs the data to the correct destination. It acts like a postal service, ensuring that your data packets reach the right address.

Interactive Activity: Let's simulate sending a message. One student will be the sender, and another will be the receiver. The sender writes a message, and the class ensures it gets to the receiver without any errors.

Open Systems Interconnection (OSI) Model

The **Open Systems Interconnection (OSI) Model** is a conceptual framework used to understand and implement standard protocols in network communication. Developed by the International Organization for Standardization (ISO) in the late 1970s, the OSI model was designed to standardize the functions of a telecommunication or computing system irrespective of its underlying structure and technology.

History and Development

When: The OSI model was developed in the late 1970s and published in 1984.

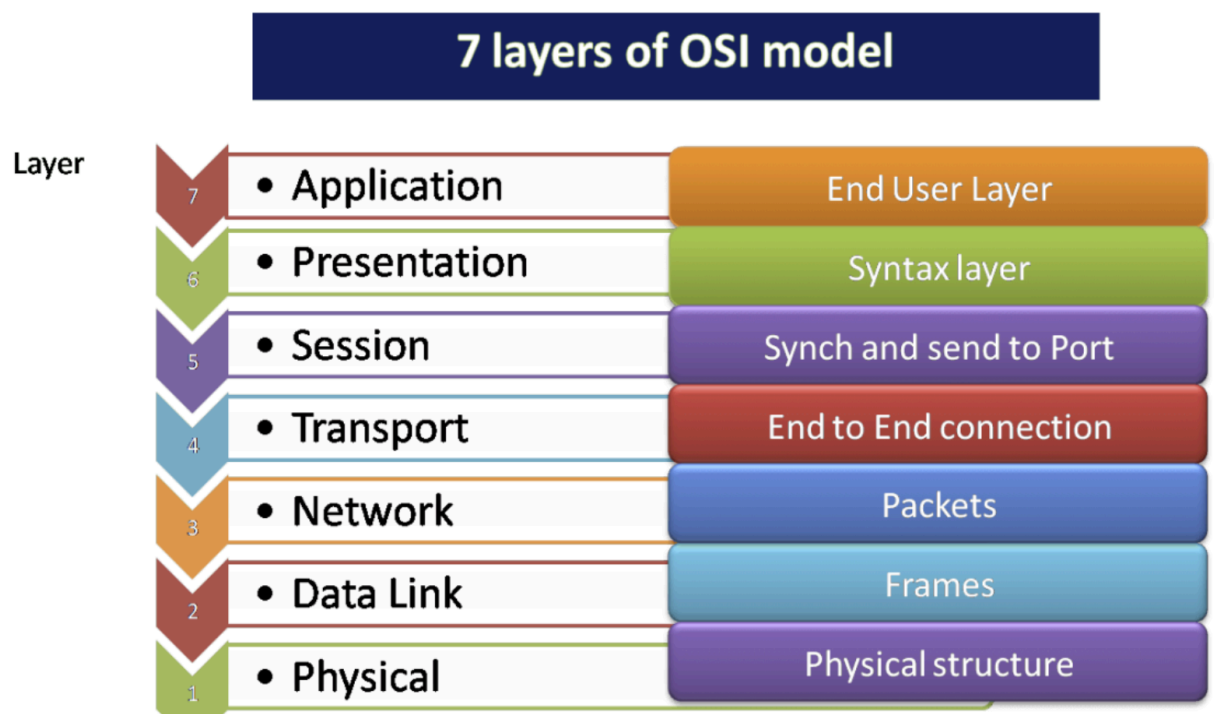
Who: The model was created by the International Organization for Standardization (ISO).

How:

- The OSI model was introduced to address the incompatibility between different networking systems and ensure that various types of network hardware and software could communicate effectively.
- It serves as a universal set of standards and protocols that enable different devices and systems to communicate over a network.

The Seven Layers of the OSI Model

The OSI model consists of seven layers, each with specific functions and protocols. These layers can be grouped into two categories: upper layers (focused on user interface and data processing) and lower layers (focused on data transport).



The Seven Layers of the OSI Model

1. **Physical Layer:**
 - Deals with the physical connection between devices and the transmission of raw bitstreams over a medium (e.g., cables, radio waves).
2. **Data Link Layer:**

- Ensures error-free data transfer between two physically connected devices; handles error detection, correction, and flow control.
- 3. **Network Layer:**
 - Manages data routing, forwarding, and addressing; determines the best path for data to travel from source to destination.
- 4. **Transport Layer:**
 - Provides end-to-end communication services; ensures complete data transfer, error recovery, and flow control.
- 5. **Session Layer:**
 - Manages and controls the connections (sessions) between computers; establishes, maintains, and terminates sessions.
- 6. **Presentation Layer:**
 - Translates, encrypts, and compresses data; ensures data is in a usable format and properly formatted for the application layer.
- 7. **Application Layer:**
 - Provides network services directly to user applications; interfaces with software that implements communication components (e.g., HTTP, FTP, SMTP).

Comparison with the TCP/IP Model

The OSI model is often compared to the TCP/IP model, which is the foundation of the Internet. The TCP/IP model has four layers: Link, Internet, Transport, and Application. Here's a brief comparison:

- **Link Layer (TCP/IP):** Combines OSI's Physical and Data Link layers.
- **Internet Layer (TCP/IP):** Corresponds to the OSI's Network layer.
- **Transport Layer (TCP/IP):** Corresponds to the OSI's Transport layer.
- **Application Layer (TCP/IP):** Combines OSI's Application, Presentation, and Session layers.

TCP/IP Model: 4 layers

1. Link
2. Internet
3. Transport
4. Application

Understanding the TCP/IP Model and Common Protocols

Imagine the Internet as a huge network of connected computers that communicate with each other. To make this work, the TCP/IP model organizes how data is sent and received. Think of it as a recipe with four key steps, each with a different job. Let's break it down and see how some common Internet protocols fit into this model.

The Four Layers of the TCP/IP Model

1. Link Layer (The Delivery Trucks)

What It Does:

- Handles the physical connection and data transport over local networks.
- Includes hardware and protocols like Ethernet and Wi-Fi.

2. Protocols/Examples:

- **Ethernet:** Used for wired connections, like connecting your computer to a router with a cable.
- **Wi-Fi:** Used for wireless connections, like connecting your phone to a wireless network.

3. Internet Layer (The Post Office)

What It Does:

- Routes data across different networks.
- Uses IP addresses to find where data needs to go and determines the best path.

4. Protocols/Examples:

- **IP (Internet Protocol):** Provides the addressing system for computers to identify and communicate with each other over the Internet.

5. Transport Layer (The Delivery Confirmation)

What It Does:

- Ensures data is delivered accurately and in the correct order.
- Manages the flow of data between devices.

6. Protocols/Examples:

- **TCP (Transmission Control Protocol):** Ensures reliable, ordered, and error-checked delivery of data. Used for applications where accuracy is crucial.
- **UDP (User Datagram Protocol):** Provides faster but less reliable data delivery. Used for applications where speed is more important than reliability.

7. Application Layer (The Web Browser)

What It Does:

- Provides network services directly to user applications.
- Handles the protocols that applications use to communicate over the network.

8. Protocols/Examples:

- **HTTP (Hypertext Transfer Protocol)**
 - **What It Does:** Used for transferring web pages on the Internet. When you visit a website, your browser uses HTTP to request and receive web pages.
 - **Example:** When you open a website like www.example.com, your browser uses HTTP to fetch the page.
- **HTTPS (Hypertext Transfer Protocol Secure)**
 - **What It Does:** Similar to HTTP but with added security. It encrypts the data exchanged between your browser and the website to keep it safe from eavesdroppers.
 - **Example:** When you enter sensitive information like passwords or credit card numbers, HTTPS ensures that this data is encrypted and secure.

- **FTP (File Transfer Protocol)**
 - **What It Does:** Used for transferring files between computers over a network. It's like sending files through a digital post office.
 - **Example:** When you upload a file to a website or download a file from a server, you might use FTP.
- **SCP (Secure Copy Protocol)**
 - **What It Does:** Used for securely copying files between computers over a network. It's similar to FTP but with added security.
 - **Example:** When you need to securely transfer a file from one computer to another, SCP ensures that the file is encrypted during transfer.
- **SSH (Secure Shell)**
 - **What It Does:** Used for securely accessing and managing remote computers. It provides a secure way to log in and execute commands on another computer over the network.
 - **Example:** When a system administrator needs to access a remote server to manage it, they use SSH to ensure the connection is secure.

Putting It All Together

Here's how it all works with a practical example:

1. **You want to visit a website** (Application Layer: Your browser uses HTTP or HTTPS to request the web page from a server).
2. **Your browser sends the request over your local network** (Link Layer: The request travels from your computer to your router).
3. **The request is routed across the Internet** (Internet Layer: The request finds the best path to the web server).
4. **The data arrives and is confirmed** (Transport Layer: TCP ensures that the web page data arrives correctly and in the right order).

Summary

- **Link Layer:** Manages physical connections and data transport locally.
- **Internet Layer:** Routes data across networks.
- **Transport Layer:** Ensures reliable data delivery.
- **Application Layer:** Handles protocols for specific applications (HTTP, HTTPS, FTP, SCP, SSH).

These protocols and layers work together to make sure that your data travels smoothly and securely across the Internet, from your computer to wherever it needs to go.

3. Browsers and How They Work

A **browser** is a software application that allows you to access and interact with websites. Popular browsers include Chrome, Firefox, Safari, and Edge. When you type a web address into a browser, it sends a request to the server, retrieves the web page data, and displays it on your screen.

Interactive Demonstration: Open a browser on the projector and show the students how a website loads by typing in a URL and explaining each step as it happens.

4. JavaScript: Bringing Websites to Life

JavaScript is a programming language that allows developers to create interactive elements on websites, like animations, forms, and dynamic content. It runs in the browser and enhances user experience by making web pages more engaging.

Interactive Code Snippet: Show a simple JavaScript example that changes the text on a web page when a button is clicked.

html

Copy code

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Example</title>
  <script>
    function changeText() {
      document.getElementById("demo").innerHTML = "Hello,
JavaScript!";
    }
  </script>
</head>
<body>
  <p id="demo">Click the button to change this text.</p>
  <button onclick="changeText()">Click Me!</button>
</body>
</html>
```

5. Synchronous vs. Asynchronous Communication

Synchronous communication happens in real-time, where one action must complete before the next one starts. For example, a phone call is synchronous because both parties are engaged at the same time.

Asynchronous communication, on the other hand, allows actions to happen independently. For example, sending an email is asynchronous because the sender and receiver do not need to be available simultaneously.

Interactive Example: Compare a phone call (synchronous) and email (asynchronous) in real life. Then, explain how JavaScript uses asynchronous communication with a simple example using `setTimeout`.

javascript

Copy code

```
console.log("First message");
setTimeout(() => {
    console.log("Second message after 2 seconds");
}, 2000);
console.log("Third message");
```

6. Servers and Clients: The Heart of Internet Communication

In the world of the Internet, there are two main players: servers and clients. A **server** is a powerful computer that provides data or services to other computers, known as **clients**. For instance, when you visit a website, your computer (the client) sends a request to the website's server, which then sends back the web page data for you to view.

Interactive Demonstration: Use a diagram to show the interaction between a client and a server when accessing a website.

7. IP Address: The Digital Address System

Every device connected to the Internet has a unique identifier called an **IP address**. It's like a home address for your computer, helping data find its way to the right place. There are two types of IP addresses:

- **IPv4:** Consists of four numbers separated by periods (e.g., 192.168.1.1).
- **IPv6:** A longer format with numbers and letters separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

Interactive Activity: Look up the IP address of your school's network or your personal device.

8. Device Identification: Recognizing Devices on the Network

Device identification involves recognizing a device based on its unique characteristics, such as its IP address or MAC address (a hardware address). This helps in tracking and managing devices on a network, ensuring that data reaches the correct device.

9. Data Routing: Finding the Best Path

When data travels across the Internet, it doesn't take a direct route. Instead, it hops from one device to another, finding the best path to its destination. This process is called **data routing**. Routers are devices that direct data along these paths, ensuring it reaches the correct destination quickly and efficiently.

Interactive Diagram: Show a visual representation of data hopping from one router to another to reach its destination.

10. Network Security: Keeping Data Safe

As data travels across the Internet, it's important to keep it safe from unauthorized access, attacks, or damage. This is where **network security** comes in. Network security involves using tools like firewalls, encryption, and antivirus software to protect data and resources on a network.

11. Domain Name System (DNS): The Internet's Phone Book

Remembering IP addresses can be challenging, which is why we use domain names like www.example.com. The **Domain Name System (DNS)** is like the phone book of the Internet. It translates human-friendly domain names into IP addresses that computers use to identify each other on the network.

Interactive Activity: Look up the IP address of a website using an online DNS lookup tool.

12. Hosts File: Local DNS Mapping

The **hosts file** is a small text file on your computer that maps domain names to IP addresses. It's a local version of a DNS lookup. When you type a domain name into your browser, your computer checks the hosts file first to see if there's an entry for that domain. If there is, it uses the IP address listed there instead of querying a DNS server.

C:\Windows\System32\drivers\etc\hosts

Interactive Demonstration: Show students how to view and edit the hosts file on their computer (with proper warnings about not making changes without understanding the consequences).